

3-MONTH SOFR FUTURES (SR3)

Spread & Curve Structure — Fundamental Reference Monetary Policy, Rates Markets, and the Forward Rate Curve

This document presents a fact-based reference on the fundamental drivers of CME 3-Month SOFR futures (ticker: SR3), with particular focus on the forces that shape the forward rate curve and the spread structure between contract months. SR3 futures are the primary exchange-traded instrument for hedging and speculating on the path of US short-term interest rates, having replaced Eurodollar futures as the market standard following the LIBOR transition completed in June 2023. The SR3 curve is the most liquid and institutionally important short-term interest rate (STIR) futures market in the world. Its spreads and flies encode the market's probability-weighted expectations about Federal Reserve monetary policy, inflation, economic growth, and financial conditions — across a horizon extending up to ten years forward. Understanding the mechanics, the data, and the analytical framework for this market is the subject of this document.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and Settlement Mechanics

3-Month SOFR futures replaced 3-Month Eurodollar futures as the benchmark short-term interest rate contract on CME following the permanent cessation of USD LIBOR on June 30, 2023. The contract structure closely mirrors Eurodollar conventions to facilitate the transition.

Parameter	Detail
Exchange	CME Group (Chicago Mercantile Exchange)
Ticker	SR3 (CME); also traded as SOFR3M on Bloomberg terminal
Underlying	3-Month Term SOFR rate (the compounded SOFR rate over the contract's reference quarter)
Contract size	$\$2,500 \text{ per basis point} \times 100 = \$2,500 \times (100 - \text{rate}) = \text{notional}$ $\$1,000,000 \times 90/360$
Quotation	100 minus the annualised 3-month SOFR rate. Example: a price of 94.75 implies a rate of 5.25%. Prices rise when rates fall; prices fall when rates rise.
Minimum tick	0.005 (half a basis point) = \$12.50 per contract for most expiries; 0.0025 for front two contracts in certain conditions
Contract months	March (H), June (M), September (U), December (Z) — quarterly cycle. Active contracts extend approximately 10 years forward (40 quarterly contracts). Serial monthly contracts also listed.
Settlement type	Cash-settled. Final settlement price = 100 minus the 3-month compounded SOFR rate for the reference period, as published by the NY Fed.
Last trading day	Third Wednesday of the contract month (same convention as Eurodollar)
Reference period	The 3-month period beginning on the third Wednesday of the contract month. The contract settles to the compounded daily SOFR fixing over that 3-month window — not a single point-in-time rate.
Open interest	Concentrated in the front 8–12 contracts (2–3 years); meaningful liquidity to approximately 20 contracts (5 years); activity thins but remains tradeable to 40 contracts (10 years)
DV01	\$25 per contract per basis point (0.01) move in rate

1.1 SOFR: The Underlying Rate

The Secured Overnight Financing Rate (SOFR) is administered by the Federal Reserve Bank of New York. It is published each business day at approximately 8:00 AM ET, representing the prior business day's rate. Understanding what SOFR measures is essential for interpreting the futures curve:

- **What SOFR measures:** SOFR is a broad measure of the cost of borrowing cash overnight collateralised by US Treasury securities (repo transactions). It is calculated as a volume-weighted median of overnight Treasury repo transactions, covering three segments: tri-party repo (cleared through DTCC), General Collateral Finance (GCF) repo, and bilateral Treasury repo cleared through FICC.
- **Daily transaction volume:** the underlying repo market transacts approximately \$1–1.5 trillion per day, making SOFR one of the most deeply anchored benchmark rates globally. This volume gives SOFR strong manipulation resistance compared to the survey-based LIBOR it replaced.
- **SOFR vs LIBOR:** LIBOR was a forward-looking term rate (e.g. 3-month LIBOR fixed the rate for the next 3 months in advance). SOFR is overnight and backward-looking — the compounded SOFR for a 3-month period is only known at the end of that period. This is a fundamental structural difference:

SR3 futures price the expected compounded overnight rate over the reference quarter, not a single forward-looking fixing.

- **SOFR fixings and spikes:** SOFR is sensitive to year-end balance sheet pressures, quarter-end regulatory reporting, and Treasury settlement surges. Large Treasury issuance that increases the supply of collateral can temporarily depress SOFR (more collateral available relative to cash). This creates predictable seasonal patterns in the SOFR fixing around quarter-end and year-end dates.

1.2 Price Quotation and Spread Arithmetic

The inverted price quotation (100 minus rate) has direct implications for how spreads are expressed and interpreted:

- **Spread = C1 price minus C2 price.** A positive spread means the front contract is priced higher than the back — i.e. rates are expected to FALL (front rate > back rate in price terms, but front rate < back rate in yield terms). This corresponds to an inverted yield curve — rates are expected to decline over the next quarter.
- **Negative spread:** front contract priced lower than back — rates expected to RISE. Corresponds to a normal (upward-sloping) curve where future rates are higher than current rates.
- **The fly:** long C1 and C3, short 2×C2. A positive fly means C2 is cheap relative to C1 and C3 — the market prices a temporary rate move at C2's reference period relative to the path on either side. Flies directly capture the market's view of whether a specific FOMC meeting will deliver a hike, cut, or hold relative to what adjacent meetings are pricing.
- **DV01 of a spread:** a 1-basis-point parallel shift in rates moves each contract by \$25. A C1-C2 spread has a net DV01 of approximately zero to parallel shifts (long and short offset) but is sensitive to the slope change between the two contract months.

2. Federal Reserve Policy — The Primary Driver

The Federal Reserve's Federal Open Market Committee (FOMC) sets the Federal Funds Rate target range, which is the primary anchor for all US short-term interest rates including SOFR. Understanding FOMC mechanics is the prerequisite for any analytical framework applied to the SR3 curve.

2.1 The Federal Funds Rate and SOFR

- **The Fed Funds Rate target range:** the FOMC sets a target range (e.g. 5.25%-5.50%) for the overnight Fed Funds rate — the rate at which banks lend reserves to each other overnight. The Fed enforces this range through two administered rates: Interest on Reserve Balances (IORB) and the Overnight Reverse Repo Facility (ON RRP).
- **IORB (Interest on Reserve Balances):** the rate the Fed pays banks on reserves held at the Fed. This sets a soft floor for overnight rates — banks will not lend below IORB since they can always earn it risk-free.
- **ON RRP (Overnight Reverse Repo Facility):** the rate the Fed offers to non-bank counterparties (money market funds, GSEs) to park cash overnight against Treasury collateral. This acts as a floor for the broader money market, since non-bank entities earning ON RRP will not lend below it.
- **SOFR's relationship to Fed Funds:** SOFR trades in the same vicinity as the Fed Funds target range but is distinct. SOFR is a secured rate (collateralised by Treasuries); Fed Funds is unsecured. Secured rates are typically slightly below unsecured rates in normal conditions, so SOFR trades near the bottom of the Fed Funds target range. This relationship holds consistently in non-stressed conditions.

2.2 FOMC Meeting Schedule and Its Direct Impact on the SR3 Curve

The FOMC meets eight times per year on a published schedule. Each meeting produces a policy decision (hold, hike, cut). The SR3 curve is, fundamentally, a chain of FOMC meeting expectations:

- Each quarterly SR3 contract covers a 3-month reference period that typically contains one or two FOMC meetings. The price of each contract reflects the expected compounded SOFR over that period — which is the expected average policy rate for that quarter.
- A 25bp FOMC rate change moves the SOFR rate by exactly 25bp for the entire quarter in which it occurs. This translates to approximately a 6.25bp move in the futures price of the contract covering a quarter with one FOMC meeting ($25\text{bp} \times 1/4$ of the year, roughly), though the exact impact depends on when in the quarter the meeting falls.
- The FOMC date relative to the start and end of the reference quarter determines how much of the quarter's compounded rate reflects the new vs old rate. A rate change at the very start of the quarter is fully priced; a change at the end affects only a few days of the compounding.

This "days in quarter" arithmetic is one of the most important mechanical features of SR3 spread and fly analysis. Two adjacent contract months can have very different sensitivities to the same FOMC meeting depending on when that meeting falls relative to each contract's reference period.

2.3 Fed Funds Futures vs SR3 Futures

Both Fed Funds futures (FF, monthly) and SR3 futures (quarterly) price FOMC expectations. Their relationship is important for spread traders to understand:

Feature	Fed Funds Futures (FF)	3-Month SOFR Futures (SR3)
Underlying rate	Monthly average overnight Fed Funds rate	Compounded daily SOFR over a 3-month reference quarter

Contract cycle	Monthly — every calendar month	Quarterly — March, June, September, December
Horizon	Active to ~3 years forward	Active to ~10 years forward; deep liquidity to 5 years
Primary use	Precise FOMC meeting probability extraction; short-horizon hedging	Curve trading, longer-horizon rate hedging, spread/fly strategies
Precision for FOMC	Higher — monthly contract isolates specific meetings	Lower for individual meetings; better for multi-meeting path
Liquidity	Good in front 12 months; thins rapidly beyond	Deepest short-rate liquidity globally; extends to 10 years
DV01 per contract	~\$41.67 per bp	\$25 per bp
Interaction	FF rates imply SOFR via the FF-SOFR basis spread	SR3 curve reflects path of Fed Funds target via basis adjustment

2.4 The FOMC Communication Framework

The FOMC communicates its intentions and assessments through multiple channels, each of which has a distinct and documented impact on the SR3 curve:

- **FOMC Statement:** released at 2:00 PM ET on each meeting day. Contains the rate decision, the vote tally, and key language about the economic outlook and future policy direction. Word-by-word changes in the statement language (e.g. "some" vs "additional" policy firming) are interpreted by markets as signals about the future rate path. The statement moves the front of the SR3 curve immediately on release.
- **Summary of Economic Projections (SEP) / "Dot Plot":** released quarterly (March, June, September, December meetings). Each FOMC participant submits their projection for the appropriate Fed Funds Rate at year-end for the next 2–3 years and in the long run. The dot plot is the most direct forward-looking signal from FOMC members about the expected rate path. Shifts in the median dot move the SR3 curve along its entire length proportionally to where the dots moved.
- **Press Conference:** held after every meeting since 2019. The Chair's prepared remarks and Q&A; responses provide nuance beyond the statement language. Spontaneous remarks have historically moved the curve more than scripted statements.
- **FOMC Minutes:** released three weeks after each meeting. Provide detailed discussion of the policy debate, including disagreements among members and discussions of specific economic conditions. Minutes tend to move the mid-curve (C3–C8) more than the front.
- **Speeches by FOMC members:** Fed Chair, Vice Chair, and regional Fed presidents give speeches continuously between meetings. Voting member speeches are more market-moving than non-voting members. The Chair's speeches at venues like Jackson Hole (August) and the Brookings Institution are particularly watched for signals about upcoming policy shifts.
- **Jackson Hole Economic Symposium (August):** the annual Federal Reserve conference in Wyoming. The Chair's speech here has historically previewed significant policy pivots — including QE tapering announcements (2021) and the commitment to sustained tightening (2022). It is the most important non-meeting communication event of the year.

3. Macroeconomic Fundamentals — The Rate-Setting Inputs

The FOMC's rate decisions are determined by its assessment of the economy against its dual mandate: maximum employment and price stability (2% inflation target). The following economic variables have documented, systematic relationships with the SR3 curve.

3.1 Inflation Data

Inflation is the most watched economic variable for the SR3 curve. The FOMC's 2% inflation target (measured by PCE) defines the terminal destination for price stability, and any deviation from this target — sustained above or below — generates a policy response that reprices the SR3 curve.

Indicator	Publisher	Frequency	Release Timing	Fed Target / Reference	Curve Impact
PCE (Personal Consumption Expenditures) Price Index	BEA	Monthly	~4 wks after month end, 08:30 ET	2% YoY — the Fed's primary inflation target	Most important: above-target = hike signal; below = cut signal; C2-C12 repricing
Core PCE (ex food & energy)	BEA	Monthly	Same release as PCE	2% YoY — closest to Fed's operating target	Core surprises move the whole SR3 curve; drives dot plot revisions
CPI (Consumer Price Index)	BLS	Monthly	~2 wks after month end, 08:30 ET	Not the Fed target but closely watched as leading PCE signal	CPI surprise is the single most market-moving monthly data event for SR3
Core CPI (ex food & energy)	BLS	Monthly	Same release as CPI	Watched for "super-core" (services ex shelter) as Fed focus metric	Super-core services CPI most relevant for assessing wage-driven inflation
PPI (Producer Price Index)	BLS	Monthly	1 day before CPI	Leading input cost signal; feeds into PCE with lag	Moves front-end SR3 as early read on next month's CPI/PCE direction
5y5y Breakeven Inflation (market-implied)	TIPS market	Daily	Continuous	Market's long-run inflation expectation; Fed monitors for "anchoring"	Rising long-run breakevens can accelerate rate hike path; affects C8-C20

3.2 Labour Market Data

The employment side of the Fed's dual mandate is assessed primarily through the following indicators:

- **Non-Farm Payrolls (NFP):** published by BLS on the first Friday of each month, 08:30 ET. Reports net job creation across all non-agricultural sectors. The single most market-moving monthly data release for SR3 after CPI. A large positive surprise (well above consensus) signals a tight labour market, reinforcing the case for higher rates — bearish for SR3 prices (rates implied higher). A large miss signals labour market softening — bullish for SR3.
- **Unemployment rate:** released simultaneously with NFP. The Fed monitors whether unemployment is at, above, or below the estimated Non-Accelerating Inflation Rate of Unemployment (NAIRU), currently estimated around 4.0-4.5%. Below NAIRU = labour market too tight, inflationary pressure. Above NAIRU = slack in the labour market, deflationary tendency.
- **Average Hourly Earnings (AHE):** released with NFP. Measures wage growth, which is a direct input into services inflation. AHE above 4% YoY in a context of 2% inflation target implies real wages rising faster than productivity allows — a sustained inflationary pressure the Fed must address.

- **JOLTS (Job Openings and Labour Turnover Survey)**: monthly, published ~5 weeks after month-end by BLS. Job openings vs unemployed workers ratio (the "V/U ratio") is a measure of labour market tightness the Fed explicitly references. High V/U = tight market; falling V/U = loosening.
- **ADP Employment Report**: private-sector payroll estimate published two days before NFP. Sets market expectations but has an inconsistent track record as an NFP predictor. Moves SR3 modestly.
- **Initial Jobless Claims**: weekly (Thursday 08:30 ET). Most timely labour market signal. Sustained rise above ~250,000–280,000 signals deteriorating labour conditions and can shift near-term SR3 rate cut expectations.
- **Employment Cost Index (ECI)**: quarterly, published by BLS. Broadest measure of labour costs (wages + benefits). The Fed has explicitly referenced ECI as a key wage inflation indicator. Quarterly timing means surprises move the SR3 curve significantly along the mid-to-back end.

3.3 Economic Activity and Growth Data

- **GDP (Gross Domestic Product)**: quarterly, published in three vintages — advance (first month after quarter end), second estimate, and third/final. The advance estimate moves SR3 most; subsequent revisions move it less. GDP above trend (~2.0–2.5% for the US) implies the economy can absorb higher rates; below trend implies rate-cut territory.
- **ISM Manufacturing PMI**: monthly, first business day of month. A reading above 50 indicates expansion; below 50 contraction. The ISM is a leading indicator of economic activity. A sustained fall below 50 (especially below 47–48) historically precedes Fed rate cuts and moves SR3 prices higher.
- **ISM Services PMI**: monthly, typically third business day. More important than Manufacturing PMI for the US economy (services ~80% of GDP). The prices-paid subcomponent is particularly watched as a leading inflation indicator.
- **Retail Sales**: monthly, released ~2 weeks after month-end by Census Bureau. Measures consumer spending — the largest component of GDP. Strong retail sales in a tight labour market signal continued demand-pull inflation pressure.
- **Housing data** (Housing Starts, Building Permits, Existing Home Sales): monthly. Housing is one of the most rate-sensitive sectors. Sustained weakness in housing from rate hikes is a leading indicator that monetary policy is having its intended effect. Shelter CPI (the largest component of headline CPI) lags actual rent changes by 12–18 months, creating a known disconnect between market rents and measured inflation.

3.4 Financial Conditions Indices

The Fed operates monetary policy through the economy's financial conditions — the combined effect of short rates, long rates, equity prices, credit spreads, and the dollar. Several indices synthesise this:

- **Goldman Sachs Financial Conditions Index (GS FCI)**: widely referenced. Tighter financial conditions (higher FCI) represent the real-economy effect of higher rates. If financial conditions tighten sharply for reasons other than Fed action (e.g. credit spread widening, equity sell-off), this does the Fed's work for it — reducing the need for rate hikes and potentially pulling forward rate-cut expectations in the SR3 curve.
- **Chicago Fed NFCI (National Financial Conditions Index)**: weekly, released Wednesdays. Covers 105 financial variables. Negative readings = accommodative conditions; positive = tight. Moves above zero historically precede economic slowdowns and Fed pivots.
- **Credit spreads** (Investment Grade and High Yield): Bloomberg US IG and HY OAS (option-adjusted spread). Wide credit spreads signal financial stress and market risk aversion, which the Fed interprets as tightening conditions — potentially allowing for fewer hikes or earlier cuts. Sudden credit spread widening moves the SR3 curve bullish (pricing fewer hikes / more cuts) in the front end.

4. The Rate Cycle and SR3 Curve Regimes

The SR3 curve takes on fundamentally different shapes in different phases of the monetary policy cycle. Recognising the current regime is the prerequisite for interpreting spread and fly behaviour correctly.

Regime	Rate Direction	Curve Shape	Spread Behaviour	Fly Behaviour	Primary Driver
Hiking cycle	Rising rates	Inverted (front > back in price; rates rise along curve in yield terms). Curve flattens as hikes priced in.	Front spreads negative (C1 < C2 in price = C1 rate > C2 rate). Back spreads approach zero as terminal rate priced.	Flies at peak of hike path convex — market uncertain about exact peak timing.	Inflation data, labour market strength, Fed hawkish rhetoric
Peak / pause	On hold at terminal rate	Flat to mildly inverted near front; back curve may begin to price cuts, creating a U-shape.	Near-term spreads near zero (no hike/cut priced); deferred spreads positive (cuts priced).	Kink at transition from hold to cut — flies around the pivot quarter are most volatile.	Inflation persistence vs cooling, unemployment trajectory, financial stability
Cutting cycle	Falling rates	Front falls in yield (rises in price); back anchored by long-run neutral rate expectations. Curve steepens.	Front spreads positive and large (C1 price > C2 in price terms = rates declining). Narrows as cuts delivered.	Flies negative near front (rate path steep downward); positive at bottom of cycle.	Labour market deterioration, inflation below target, financial stress events
Post-cycle / normalisation	Approaching neutral	Flat or mildly upward sloping in yield terms. The terminal / neutral rate defines back curve anchor.	Spreads small and two-sided — market uncertain about direction.	Flies volatile around the neutral rate estimate — small data surprises move them significantly.	Neutral rate estimates (r^*), productivity growth, demographic trends

4.1 The Neutral Rate (r^*) — The Anchor of the Back Curve

The neutral rate (r^*) is the theoretical level of the Fed Funds Rate at which monetary policy is neither stimulative nor restrictive — consistent with the economy growing at potential with stable inflation. It is the primary anchor for pricing the long end of the SR3 curve:

- The Fed does not publish an official r^* estimate, but FOMC participants submit their "longer run" rate projection in the SEP. The median longer-run dot has historically ranged from 2.25% to 2.75%, though estimates have shifted higher since 2022 as evidence of a structurally higher r^* has accumulated.
- The Holston-Laubach-Williams (HLW) model, published quarterly by the NY Fed, is the most widely referenced academic r^* estimate. Revisions to the HLW model output move the SR3 back curve because they update the market's estimate of where rates will eventually settle in the long run.
- Factors associated with a higher r^* : stronger trend productivity growth, fiscal deficits requiring higher real rates to attract funding, demographic shifts reducing the saving glut, and increased public and private capital investment needs (green transition, AI infrastructure, defense).
- Factors associated with a lower r^* : demographic aging increasing desired saving, global savings glut (large Asian reserve accumulation), slow trend productivity growth, and high debt levels that make economies sensitive to higher rates.

4.2 The Term Premium

Beyond the expected path of short rates, long-dated SR3 contracts embed a term premium — compensation for the risk that rates deviate from expectations over long horizons:

- When term premium is high, the SR3 back curve is steeper in yield terms (higher rates further out) than pure rate expectations would imply. This typically occurs during periods of high inflation uncertainty, large fiscal deficits (increasing Treasury supply), or reduced central bank balance sheet (QT reducing demand for bonds).
- When term premium is low or negative (as it was from 2014–2021), the back curve is anchored near or below r^* estimates despite the uncertainty about the long-run rate path.
- The Adrian-Crump-Moench (ACM) term premium model, published daily by the NY Fed, decomposes the 10-year Treasury yield into rate expectations and term premium components. This is useful for SR3 back-curve positioning: if the back curve is steep primarily due to elevated term premium (not rate expectations), it may compress as uncertainty resolves without requiring a fundamental change in the rate path.

5. Key Data Events and the SR3 Market Calendar

The SR3 curve responds to a well-defined calendar of data releases and policy events. Understanding the relative importance of each and its typical zone of curve impact is fundamental to spread positioning.

Event / Release	Publisher	Frequency	Timing	Primary Curve Impact	Magnitude
CPI (headline + core)	BLS	Monthly	~2nd week, 08:30 ET	All curve — largest single monthly event	★★★★★
Non-Farm Payrolls + Unemployment	BLS	Monthly	1st Friday, 08:30 ET	C1-C8 — front more than back	★★★★★
FOMC rate decision + statement	Federal Reserve	8× per year	2:00 PM ET	All curve — immediate repricing	★★★★★
Chair press conference	Federal Reserve	8× per year	2:30 PM ET	All curve — often moves more than statement	★★★★
SEP / Dot Plot	Federal Reserve	4× per year (Mar/Jun/Sep/Dec)	2:00 PM ET with FOMC	C4-C20 — dot shifts move back curve significantly	★★★★
PCE Price Index (core)	BEA	Monthly	~4th week, 08:30 ET	C2-C12 — confirms or revises CPI signal	★★★★
FOMC Minutes	Federal Reserve	8× per year, 3-wk lag	2:00 PM ET	C3-C8 — nuance on rate path debate	★★★
ISM Services PMI (prices paid)	ISM	Monthly	~3rd business day	C1-C4 — services inflation leading signal	★★★
GDP Advance Estimate	BEA	Quarterly	~4 wks after quarter end, 08:30 ET	C2-C8 — growth trajectory signal	★★★
Retail Sales	Census	Monthly	~2nd week, 08:30 ET	C1-C4 — demand / consumer activity	★★★
Average Hourly Earnings	BLS	Monthly	With NFP, 08:30 ET	C2-C8 — wage inflation signal	★★★
Initial Jobless Claims	DOL	Weekly	Thursday, 08:30 ET	C1-C3 — most timely labour signal	★★
JOLTS Job Openings	BLS	Monthly	~5 wks after month end	C2-C6 — labour tightness / V-U ratio	★★
PPI	BLS	Monthly	Day before CPI, 08:30 ET	C1-C3 — leading cost signal	★★
Employment Cost Index (ECI)	BLS	Quarterly	Last day of Jan/Apr/Jul/Oct	C2-C8 — broad wage cost signal	★★★
Jackson Hole Speech (Chair)	Federal Reserve	Annual (August)	~10:00 AM ET	All curve — major pivot signals have occurred here	★★★★★
Fed balance sheet (H.4.1)	Federal Reserve	Weekly	Thursday, ~16:30 ET	C4-C20 — QE/QT trajectory	★★
NY Fed SOFR publication	NY Fed	Daily	~08:00 ET	C1 spot fix — basis for front convergence	★

5.1 The CPI Release — Anatomy of the Most Important Event

CPI is consistently the single most market-moving scheduled event for the SR3 curve. The following breakdown illustrates how the release propagates through the curve:

- **Headline CPI surprise vs consensus:** a miss of +0.1% (e.g. 0.4% actual vs 0.3% expected) on the monthly print typically moves the SR3 front contracts 3–8bp in yield terms within the first minute of release. The move is not proportional — it is asymmetric toward surprise.
- **Core CPI and "super-core" (services ex shelter):** as the Fed has focused on persistent services inflation, the non-housing services component (super-core) has received disproportionate attention. A hot super-core print steepens the SR3 curve in the near term (delays cuts) more than a hot energy-driven headline number.
- **Shelter CPI lag:** the shelter component of CPI (rent of primary residence + owners' equivalent rent) lags market rents by 12–18 months due to the way it is measured. Sophisticated market participants adjust for this lag: when market rents (Zillow Observed Rent Index, Apartment List data) are falling, the market prices shelter CPI cooling in future months — a factor that moderates bearish SR3 positioning even when current CPI prints remain elevated.
- **Curve response distribution:** front contracts (C1–C4) move most on CPI surprises. Mid-curve (C4–C12) moves approximately 50–70% of the front move. Back curve (C12+) moves least, anchored by long-run rate expectations that are less sensitive to single data points.

5.2 Non-Farm Payrolls — The Employment Event

- NFP day follows a specific intraday sequence: 08:30 ET release moves SR3 immediately. The magnitude depends on: (1) the headline number vs consensus, (2) prior month revisions (often as important as the headline), and (3) average hourly earnings. A strong jobs number with weak wage growth is less hawkish than the same headline with strong wages — the SR3 curve responds to the full employment report, not just the headline.
- The unemployment rate change often has an outsized impact relative to its economic significance because it is a clean, easily comparable number that drives headlines and algo-driven trading. A tenth-of-a-percent rise in unemployment toward or above the Sahm Rule threshold (0.5pp rise in 3-month average vs prior year low) has historically been associated with recession and triggers aggressive SR3 buying (rates lower).

6. Spread and Fly Mechanics on the SR3 Curve

6.1 What a Spread Expresses

An SR3 calendar spread (e.g. long Sep25 / short Dec25) is a position on the change in the 3-month rate between the two reference periods. It is not a position on the level of rates — both legs move with parallel shifts; the spread captures the slope:

- **Positive spread (front > back in price):** the market implies rates will FALL between the front and back reference quarters. The trader who is long a positive spread profits if rates fall more than priced, or if the existing rate cut is delivered as expected.
- **Negative spread:** rates expected to RISE. Long a negative spread = positioned for more rate hikes than currently priced.
- **Spread size in rate terms:** a spread of 0.25 price points = 25bp of expected rate change between the two reference quarters. If the Dec25-Mar26 spread is 0.25 (price), the market implies approximately a 25bp rate cut between Q4 2025 and Q1 2026.
- **FOMC meeting attribution:** each spread can be attributed to the FOMC meetings that fall in the gap between the two reference quarters. The spread is effectively pricing the probability-weighted expected rate change from those meetings.

6.2 What a Fly Expresses

- A fly (e.g. long Jun25 + long Dec25 – 2× Sep25) is a position on the curvature at the Sep25 point. A positive fly means Sep25 is cheap relative to Jun25 and Dec25 — the market implies a temporary rate move centred on the Sep25 quarter.
- **FOMC meeting fly:** the most common application. If a specific FOMC meeting falls entirely within the Sep25 reference period, the Sep25 fly captures the market's uncertainty about whether that meeting delivers a cut vs hold. A fly that is zero implies the market prices a smooth rate path; a wide fly implies a high-probability discrete event at that contract.
- **Convexity vs curvature:** a fly can also reflect convexity in the rate path — the non-linearity that arises because the forward rate distribution is not symmetric (rates are bounded at zero, and uncertainty compounds). Long flies have positive gamma to rate volatility: if rates move more than expected, the fly profits regardless of direction.
- **Risk-reversal flies:** when the fly is asymmetric (e.g. the long wing is in a different contract size), the position expresses a directional view on whether the kink in the curve resolves up or down.

6.3 Curve Zone Characteristics

Zone	Contracts (approx.)	Dominant Driver	Key Signals	Typical Spread Behaviour	Volatility
Ultra-front	C1-C2 (current quarter)	Known SOFR fixings, quarter-end dynamics, next FOMC meeting	Daily NY Fed SOFR publication, overnight repo market levels, money market fund flows	Spread nearly zero or single-meeting-sized (25bp if one cut priced)	Low (mechanics-driven)
Front	C2-C4 (~6-12 months)	Near-term FOMC path; CPI and NFP surprises dominate	CPI monthly releases, NFP, FOMC statement language, Fed Chair speeches	Most volatile spreads; repriced most aggressively on data surprises	Highest
Belly	C4-C8 (~12-24 months)	Terminal rate expectations; dot plot; economic growth trajectory	SEP dot plot, GDP, ISM PMIs, ECI, Jackson Hole	Moves with SEP revisions; sensitive to recession probability shifts	High

Mid	C8-C16 (~24-48 months)	Neutral rate (r^*) estimates; the pace of return from terminal to neutral	HLW model publications, Fed balance sheet trajectory (QT pace), fiscal deficit	Trends slowly; large moves on structural reassessment of r^*	Moderate
Back	C16-C40 (~48-120 months)	Long-run r^* + term premium; fiscal outlook; energy transition capex cycle	ACM term premium model, CBO fiscal projections, IEA/EIA long-run scenarios	Most anchored; moves on regime changes, not single data points	Lowest

6.4 Quarter-End and Year-End Dynamics

SOFR is consistently affected by balance sheet pressures around quarter-end and year-end dates. These are predictable, calendar-visible patterns that directly affect SR3 front contract pricing:

- **Quarter-end repo market tightening:** at each quarter end (March 31, June 30, September 30, December 31), large banks reduce their repo market activity to shrink balance sheets for regulatory reporting (Supplementary Leverage Ratio — SLR). This temporarily reduces the supply of cash in the repo market, pushing SOFR higher for a few days.
- **Year-end "turn":** the most pronounced quarter-end effect. Banks reduce repo book size most aggressively at December 31. SOFR can spike materially above its normal range in the last few business days of December. The CME SR3 December contract that covers this period prices this "turn premium" explicitly — the implied rate for the December contract is consistently higher than adjacent months to reflect this predictable spike.
- **Treasury settlement surges:** when the US Treasury settles large volumes of new issuance (particularly 3-month T-bills), the settlement absorbs cash from money markets and temporarily increases the demand for repo borrowing — pushing SOFR higher. Large auction settlement dates are observable in advance via Treasury's published auction calendar.
- **ON RRP usage as a liquidity signal:** the outstanding balance of the Fed's Overnight Reverse Repo Facility (published daily) is a real-time measure of excess liquidity in the system. When ON RRP balance falls toward zero, reserves are scarce enough that SOFR may disconnect from the IORB floor — a signal that QT has progressed to a point of potential rate pressure.

Quarter-end SOFR spikes are priced into the front SR3 contract that covers the quarter-end date. The market creates a "turn premium" that is quantifiable and predictable in direction, though variable in magnitude.

7. Fed Balance Sheet — QE, QT, and Reserves

The Federal Reserve's balance sheet directly affects the supply of bank reserves and the functioning of money markets. Since the introduction of Quantitative Easing (QE) in 2008–09, the balance sheet has become a second monetary policy instrument alongside the Fed Funds target rate. Its trajectory affects the SR3 curve primarily through two channels: reserve supply (affecting the floor for overnight rates) and term premium (affecting the back end of the curve).

7.1 Quantitative Easing (QE)

- QE involves the Fed purchasing long-dated Treasuries and agency mortgage-backed securities (MBS) in the open market, paying for them by crediting bank reserves. This increases both the size of the Fed's balance sheet and the quantity of bank reserves in the system.
- The primary effect of QE on SR3: by suppressing long-term Treasury yields (the Fed is buying them), QE compresses the term premium embedded in long-dated SR3 contracts. It also signals that the Fed is committed to accommodative policy, which anchors front SR3 contract prices near the zero lower bound.
- QE purchase pace announcements ("tapering") are among the most market-moving events for the back of the SR3 curve. The "taper tantrum" of May 2013 (when Bernanke signalled the possibility of tapering QE purchases) caused a sharp sell-off in the back of the rates curve that is the canonical example.

7.2 Quantitative Tightening (QT)

- QT is the reverse process: the Fed allows maturing securities to roll off its balance sheet without reinvestment (or, more rarely, actively sells securities). This reduces bank reserves and bank balance sheets.
- The pace of QT is announced by the FOMC in terms of monthly caps on the runoff of Treasuries and MBS separately. Changes to the QT pace (accelerating or slowing) affect the back SR3 curve through the term premium channel.
- The critical question in QT is the level of "ample reserves": the minimum level of reserves at which the Fed's rate control mechanism continues to function normally. As QT reduces reserves toward this level, repo market pressures can emerge — as occurred in September 2019 when repo rates spiked sharply (the "repo crisis"), forcing the Fed to inject reserves via repo operations. The ON RRP balance and reserve levels (H.4.1 weekly data) are closely watched as QT progresses for early warning signals.

7.3 Bank Reserves and the Transmission Mechanism

- In the current "ample reserves" framework (post-GFC), the Fed controls rates via IORB and ON RRP rather than through the scarcity of reserves. This is fundamentally different from the pre-2008 framework where the Fed controlled rates by adding or draining small quantities of scarce reserves.
- The practical consequence for SR3 spreads: as long as reserves remain "ample," the floor for overnight rates is IORB, and SOFR should remain near the bottom of the Fed Funds target range. If reserves fall below the ample threshold, the floor becomes uncertain — a risk premium is added to front SR3 contracts.

8. Cross-Market Relationships

8.1 SR3 and the US Treasury Curve

SR3 futures and the US Treasury yield curve are intimately related but distinct. Understanding the relationship is necessary for positioning across the two markets:

- **2-Year Treasury yield vs SR3 front curve:** the 2-year Treasury is the most rate-sensitive Treasury benchmark and is largely determined by the expected path of the Fed Funds rate over the next two years — essentially the same information as the front eight SR3 contracts. The 2-year yield and the equivalent SR3 strip should price the same rate path, adjusted for term premium and the Treasury-SOFR basis.
- **The Treasury-SOFR basis:** Treasuries yield slightly more than SOFR-implied rates of equivalent maturity because Treasuries carry liquidity and credit premiums (Treasury bonds are not the same as compounded overnight SOFR). This basis varies with market conditions and affects cross-market hedging calculations.
- **Yield curve inversion and SR3:** when the 2Y-10Y Treasury spread inverts (2-year yields above 10-year yields), it implies the market expects rate cuts in the medium term — which should be visible in the SR3 curve as positive spreads in the belly (4-8 contracts). Deep inversions historically precede recessions and are associated with aggressive cutting cycles that create large positive SR3 spreads as the cycle unfolds.

8.2 SR3 and Equity Markets

- Equity prices and SR3 futures are negatively correlated in most regimes: a significant equity market decline signals economic stress or recession risk, which pulls forward rate cut expectations and is bullish for SR3 (prices up = rates down).
- The "Fed put" concept: the market has historically expected the Fed to cut rates in response to severe equity market declines, creating an asymmetric support for SR3 in extreme sell-off scenarios. However, this expectation was disrupted in 2022 when the Fed continued hiking despite a significant equity bear market, demonstrating that the inflation regime determines whether the Fed put is operative.
- VIX (equity volatility) spikes are associated with SR3 front-end buying (rate cut pricing). However, extreme stress scenarios can create the opposite effect: if financial stress causes concerns about the functioning of Treasury and repo markets (as in March 2020), SOFR itself can spike temporarily even as the market prices aggressive rate cuts.

8.3 SR3 and Credit Markets

- **Investment Grade (IG) and High Yield (HY) spreads:** widening credit spreads signal financial tightening and recession risk, pulling forward rate cut expectations — bullish for front SR3. This relationship is particularly strong during financial stress events (GFC 2008-09, COVID March 2020, SVB March 2023).
- **Secured vs unsecured spreads:** the spread between SOFR (secured) and Fed Funds (unsecured) or between SOFR and commercial paper rates measures counterparty/credit stress in the money market. Widening of this spread during stress periods (as in September 2008) signals acute funding stress that can override normal monetary policy transmission.
- **Commercial real estate (CRE) credit stress:** as a significant part of bank loan books, CRE deterioration (post-COVID office vacancy, post-rate-hike cap rate compression) affects bank health and, indirectly, the credit spread / financial conditions channel that influences FOMC policy and SR3 mid-curve pricing.

8.4 SR3 and the US Dollar (DXY)

- Higher US short rates (lower SR3 prices) typically support the USD by increasing USD-denominated asset returns relative to foreign alternatives. This relationship is consistent but not one-for-one — other factors (growth differentials, risk appetite) also drive DXY.
- Conversely, when the Fed is cutting rates (rising SR3) while other central banks are still hiking or on hold, the USD weakens, which can have second-order effects on US import prices and inflation expectations — a feedback loop that the Fed monitors.
- Extreme USD strength, by tightening global financial conditions (emerging market debt serviced in USD becomes more expensive), can itself create financial stress events that pull forward Fed rate cuts — an SR3 bullish feedback loop.

9. Global Context — Other Central Banks and Cross-Currency

SR3 futures price US rates, but they do not exist in isolation. The rate decisions of other major central banks create cross-currency differentials that affect capital flows, the dollar, and US financial conditions — all of which feed back into the SR3 curve.

Central Bank	Policy Rate / Instrument	Rate Futures Equivalent	Key Interaction with SR3
European Central Bank (ECB)	Deposit Facility Rate (DFR)	ESTR / EURIBOR futures (ER3)	Fed-ECB rate differential drives EUR/USD, which affects US import prices and financial conditions. ECB pivot ahead of Fed or behind has historically moved SR3 belly.
Bank of England (BoE)	Bank Rate	SONIA futures (SO3)	UK-US rate differential; GBP/USD. BoE is often seen as a leading indicator of Fed direction in similar inflation environments.
Bank of Japan (BoJ)	Policy Rate / YCC target	TONA OIS / JGB futures	BoJ YCC adjustment / normalisation causes Japanese institutional investors to repatriate capital, increasing JGB demand and reducing USD-denominated asset demand — bear steepens US curve and moves SR3 back end.
People's Bank of China (PBoC)	Loan Prime Rate / MLF rate	No liquid futures equivalent	China easing/tightening affects global growth expectations and commodity prices, which flow through to US inflation expectations and SR3 mid-curve.
Reserve Bank of Australia (RBA)	Cash Rate	BBSW / ASX 30-day interbank futures	AUD/USD and commodity price proxy. RBA cycle is often 3–6 months ahead of or behind the Fed; used as a forward-looking comparison.

9.1 The Bank of Japan — The Most Important Cross-Currency SR3 Driver

Japanese institutional investors (life insurers, pension funds, regional banks) hold substantial USD-denominated assets and are a significant buyer of long-dated US Treasuries. BoJ policy changes affect these flows in documented ways:

- When BoJ maintains ultra-low yields (YCC), Japanese investors are incentivised to invest overseas for higher returns, supporting demand for US Treasuries and compressing US term premium — bullish for the SR3 back end.
- When BoJ raises rates or adjusts YCC bands (as occurred July and October 2023, March 2024), Japanese domestic bonds become more attractive, potentially triggering repatriation from USD assets. This reduces demand for US Treasuries and pushes up US term premium — bearish for SR3 back contracts.
- Currency hedging costs also matter: when USD-JPY is high and the BoJ-Fed rate differential is large, the cost of hedging USD exposure back to JPY (via FX forward or cross-currency basis swap) makes unhedged USD investment relatively more attractive. Changes in the cross-currency basis (USD-JPY CRS) directly affect Japanese demand for US fixed income.

9.2 Fiscal Policy and Treasury Supply

US fiscal deficits require Treasury issuance to finance them. The volume, maturity, and pace of Treasury supply affects the SR3 back end through the term premium channel:

- Large Treasury deficits increase the supply of bonds that must be absorbed by the market. When the Fed is engaged in QT (reducing balance sheet), the private sector must absorb all new issuance plus

the runoff from the Fed's portfolio — a "double tightening" of supply that pushes up term premium and back-end SR3 implied rates.

- Treasury's Quarterly Refunding Announcements (QRA), published by the Treasury Borrowing Advisory Committee (TBAC), outline the expected issuance volumes and maturity composition for the coming quarter. Larger-than-expected issuance is bearish for the SR3 back end; smaller-than-expected is bullish.
- The Congressional Budget Office (CBO) publishes long-run deficit projections annually. Revisions that imply structurally larger deficits shift the SR3 back end toward higher implied rates through both the term premium and long-run r^* channels.

10. Summary Data Release Calendar

Release	Publisher	Frequency	Typical Timing	Primary SR3 Curve Impact
CPI (headline + core)	BLS	Monthly	~2nd week, Tue/Wed, 08:30 ET	All curve — largest monthly event; biggest surprise potential
Non-Farm Payrolls	BLS	Monthly	1st Friday, 08:30 ET	C1-C8 — employment side of mandate
Average Hourly Earnings	BLS	Monthly	With NFP, 08:30 ET	C2-C8 — wage inflation signal
PCE Price Index (core)	BEA	Monthly	~4th week, 08:30 ET	C2-C12 — the Fed's primary inflation target
FOMC Rate Decision + Statement	Federal Reserve	8× per year	2:00 PM ET	All curve — immediate repricing of path expectations
FOMC Press Conference	Federal Reserve	8× per year	2:30 PM ET	All curve — often largest single intraday move
SEP / Dot Plot	Federal Reserve	4× per year	2:00 PM ET with Mar/Jun/Sep/Dec FOMC	C4-C20 — path revision; back end most affected by r^* dot
FOMC Minutes	Federal Reserve	8× per year, 3-wk lag	2:00 PM ET	C3-C8 — debate nuance; less than statement/presser
Jackson Hole Speech	Federal Reserve	Annual (August)	~10:00 AM ET	All curve — historically associated with major pivots
GDP Advance Estimate	BEA	Quarterly	~4 wks after quarter end, 08:30 ET	C2-C8 — growth trajectory
Initial Jobless Claims	DOL	Weekly	Thursday, 08:30 ET	C1-C3 — most timely weekly labour signal
JOLTS Job Openings	BLS	Monthly	~5 wks after month end, 10:00 ET	C2-C6 — V/U ratio; labour tightness
ISM Manufacturing PMI	ISM	Monthly	1st business day, 10:00 ET	C1-C4 — activity leading indicator
ISM Services PMI (prices paid)	ISM	Monthly	~3rd business day, 10:00 ET	C1-C4 — services inflation signal
Employment Cost Index (ECI)	BLS	Quarterly	Last day of Jan/Apr/Jul/Oct, 08:30 ET	C2-C8 — broad wage signal; quarterly timing means large surprise potential
PPI	BLS	Monthly	Day before CPI, 08:30 ET	C1-C3 — leading input cost indicator
Retail Sales	Census	Monthly	~2nd week, 08:30 ET	C1-C4 — consumer demand signal
Treasury QRA (TBAC)	US Treasury	Quarterly	~1st week of Feb/May/Aug/Nov	C8-C40 — issuance supply signal for back end
Fed Balance Sheet (H.4.1)	Federal Reserve	Weekly	Thursday, ~16:30 ET	C4-C20 — QT pace; reserve level monitoring
NY Fed ON RRP balance	NY Fed	Daily	~07:00 ET	C1 — excess liquidity / floor pressure signal
NY Fed SOFR Publication	NY Fed	Daily	~08:00 ET	C1 — daily fixing; front contract basis and convergence

HLW Neutral Rate (r^*) Model	NY Fed	Quarterly	~6 wks after quarter end	C16-C40 — long-run rate anchor revision
CBO Budget and Economic Outlook	CBO	Annual + updates	January + August	C16-C40 — fiscal deficit path; term premium implications

3-Month SOFR futures represent the most institutionally complete and analytically demanding spread market covered in this series. Unlike commodity markets where physical supply and demand can be measured directly, the SR3 curve is a probability-weighted integral of policy expectations across all possible macroeconomic paths. Each contract month prices the expected compounded overnight rate for a specific 3-month window — which is the output of a FOMC reaction function applied to economic data that has not yet been released. The curve is, in this sense, the market's collective macroeconomic model.

The edge in SR3 spread and fly trading comes from three sources: a more precise understanding of the mechanics (how FOMC meeting dates fall within contract reference periods, quarter-end dynamics, the "days in quarter" arithmetic); a more accurate macro forecast (better inflation or employment projections than the consensus embedded in current prices); and a better assessment of the risks around the consensus (the asymmetry of outcomes, which flies are mispriced relative to the actual probability distribution of policy paths). All three require systematic, data-driven analysis — not narrative-driven views.